AI-based Village Pond Planning System — Technical Report

Author: Rahul Raj · **Course:** 7th semester, Assignment 1 · **Submission:** 5 September 2026 **Repository:** https://github.com/Rahul5977/AI-BasedPondAnalysis **Phase 2 route:** `POST /api/v1/analyzeContour` — public URL via `make tunnel` (ngrok) during the demonstration; local `http://localhost:8000/api/v1/analyzeContour`.

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1. Problem statement and objectives

Water conservation in rural India depends on small ponds built where the terrain will actually deliver runoff to them. Choosing such a site by hand needs a topographic survey, a rainfall record, a runoff estimate and a storage calculation — expertise a Gram Panchayat rarely has on hand. The assignment asks for a web application that does this from geospatial and satellite data and presents the result on an interactive map (FR1–FR8).

The objective of this implementation is narrower and sharper than "a map with numbers": **every figure must be derived from the uploaded input, carry its unit and an honest uncertainty band, and be traceable to a named, cited algorithm** that the author can defend live. The examined topic — how the area is selected and how the catchment is computed — is therefore where most of the engineering depth sits (§4–§6).

2. Requirements coverage

FR	Requirement	Where it is met	Evidence
1	Satellite imagery for a selected village	Esri World Imagery basemap; village derived from the upload and named by reverse-geocoding its centroid; focus mask dims the outside	`docs/figures/p1-walking-skeleton.jpg`
2	Contour map visualisation	Marching-squares contours from the working DEM, 1/2/5/10 m, labelled, Douglas-Peucker simplified (5 097 → 1 408 vertices at 2 m)	`p2-contours-5m.jpg`
3	Land available for excavation	Specification-pattern constraints on slope, water buffer (NDWI), habitation band, land cover, ownership, contiguity; parcels as polygons	`p4-ndwi-opencv.png`, `available_land` route
4	Catchment of a selected point	Priority-Flood + D8 + nearest-channel snapping + upstream BFS; polygon, area, longest flow path, relief; validated	`p2-click-to-catchment.jpg`, §7
5	Historical rainfall from public APIs	Open-Meteo ERA5-Land (45 yr) with NASA POWER fallback; Weibull 75 % dependable, JJAS share, rainy days, Gumbel 25-yr	`p3-rainfall-panel.jpg`
6	Runoff volume	SCS-CN on the daily series, rational coefficient, Strange's table — a range with its spread	`p3-design-panel.jpg`
7	Pond depth and storage	Cost-optimised excavated frustum, EAV curve, daily water balance → fill reliability, spillway, BoQ, confidence	`p3-design-panel.jpg`

FR	Requirement	Where it is met	Evidence
8	All results overlaid	Results overlay with the six PDF items, pond footprint drawn at the outlet, 13 toggleable layers	`p5-all-overlays.jpg`

3. System architecture

3.1 Layers and services

The application is layered (ADR 0001) with dependencies pointing inwards only, enforced by `tests/test_layering.py`: `api` (routers, zero business logic) → `engines` (pure numpy algorithms and use-case workflows) → `providers` (external data adapters) / `repositories` (persistence) → `domain` (units, errors, value objects; imports nothing but numpy).

Eleven Docker Compose services, each arriving in the phase that first called it (ADR 0004): PostGIS, Redis, MinIO, the API, two Celery workers (interactive/heavy bulkheads), beat, TiTiler, nginx + the React/MapLibre SPA, Prometheus, Grafana.

3.2 Stack reconciliation against the suggested stack

Suggested	Chosen	Why (ADR)
Flask or FastAPI	FastAPI	Typed contract generated for free (OpenAPI, 40+ operations), async job routes, dependency injection for ports (ADR 0005)
MongoDB / PostgreSQL	PostgreSQL + PostGIS	Boundaries, `ST_Equals` village reuse, spatial types; append-only audit rules in the database (ADR 0006)
Elevation APIs (point queries)	Full DEM rasters from the uploaded contour map (and a documented provider-tile adapter)	Catchment delineation needs a grid, not points; point APIs cannot give flow direction (ADR 0007, 0011)
Rainfall APIs (IMD, Open-Meteo, NASA POWER)	Open-Meteo ERA5-Land primary, NASA POWER fallback	Keyless, daily, 45 years in 2 s; IMD gridded needs registration (decision log)
OpenCV	OpenCV for the NDWI water mask: Otsu threshold, morphological open/close, connected components	Used where it is the right tool, not decoratively (ADR 0017)
A basic frontend library	React + MapLibre GL	Vector + raster layers, offline-first service worker (ADR 0018)

3.3 Design patterns applied

Pattern	Where	Purpose
Ports & Adapters	`DEMProvider`, `RainfallProvider`, `ObjectStore`, `JobRunner`, repositories	Same pipeline in Docker, CI and tests; no mocks (ADR 0013)
Strategy	`RunoffMethod` (SCS-CN, rational, Strange); `DEMProvider` implementations	A range instead of one number; extensibility evidence (ADR 0010)
Specification	`SlopeUnder & ~WithinBuffer & …`	Readable, self-naming land constraints (ADR 0017)

Pattern	Where	Purpose
Builder	`PondDesignBuilder`	Stage-by-stage assembly of the FR7 payload (ADR 0016)
Decorator	`Retry ∘ CircuitBreaker ∘ Cached`	Resilience without touching the adapters
Chain of Responsibility	`FallbackChain`, ordered elevation strategies in the KML parser	First provider/strategy that answers wins; provenance recorded
Flyweight	Curve-number table keyed by (land cover, soil group)	Eleven classes × four groups, not a raster of floats
Observer	WebSocket job progress over the job row	One source of truth, pushed on change
Saga	Persistence steps of the contour pipeline	Compensations in reverse on failure (ADR 0019)
State machine	Recommendation lifecycle	Illegal transitions are impossible, not merely unlikely
Transactional outbox	Recommendation events → audit log	Same-transaction event, asynchronous projection
Bulkhead	Two Celery queues and worker pools	A heavy job cannot delay a click

4. Methodology

4.1 From contours to a DEM (P1)

The uploaded KML/KMZ is parsed tolerantly (namespace-agnostic, `<Folder>` root accepted) but elevation is read strictly, by an ordered strategy — Z coordinate, then a whitelisted `ExtendedData` field, then the placemark name — with numeric `ID` fields rejected, because the sample carries exactly such a decoy (ADR 0011). The UTM zone is derived from the file's centroid; `assert_crs()` refuses any computation in degrees.

Contour vertices are densified and triangulated (Delaunay TIN, linear interpolation), then lightly smoothed (Gaussian, σ = 1 cell). The grid resolution is *derived*: cell = (hull area / total contour length) / 4, floored at the source DEM's resolution, which is inferred from the file's own metadata by an evidence-weighted match against a table of known datasets (the sample names SRTM five times and by raw filename; a generic blurb names Copernicus once). On the sample this gives a 30 m grid of 110 × 89 cells, and the honest statement that a 1 m contour interval from a 30 m source is interpolated precision, not measured accuracy.

4.2 Hydrological conditioning and flow routing (P2)

- **Depression filling and flat resolution:** Priority-Flood + ε (Barnes, Lehman & Mulla 2014; Wang & Liu 2006). One pass, O(n log n); the fill-depth raster is itself a figure (`p2-sink-fill-before-after.png`: 2 141 cells, max 8.6 m — TIN hollows and a river floor below its map-edge exit).
- **Flow direction:** D8 (O'Callaghan & Mark 1984), steepest descent over eight neighbours, diagonals divided by √2 (ADR 0009: deterministic, the GIS default, defensible; D-∞ is a documented stub).
- **Flow accumulation:** one descending-elevation pass — on a filled+ε surface that order is topological, so a single sorted loop is exact. Pure numpy, no numba.
- **Streams:** cells with ≥ 5 ha upstream (a threshold in area, so it means the same on any grid), traced into links with Strahler (1957) order; on the sample 101 links, order 3, 24.4 km, visibly following the Sivnath and its tributaries (`p2-streams-on-satellite.jpg`).

- **Derived surfaces:** Horn (1981) slope/aspect/hillshade; Zevenbergen & Thorne (1987) profile and plan curvature (ArcGIS sign convention); TWI = $\ln(a / \tan \beta)$ (Beven & Kirkby 1979).

4.3 Catchment delineation (FR4)

A click one cell off a channel returns a hillslope catchment orders of magnitude too small, so the pour point is snapped to the **nearest** cell within 150 m that drains ≥ 2 ha — nearest, not largest, so a tributary click is not dragged onto the main river — and the distance moved is returned and shown. The catchment is the breadth-first search over the inverse D8 graph from the snapped outlet; area, perimeter, longest flow path (BFS distances), relief and outlet elevation follow. A catchment reaching the map edge is flagged `catchment_truncated`. Uncertainty on the area is stated as max(15 %, one cell ring around the perimeter) — on a 38 ha catchment of 30 m cells that is ±26 %.

4.4 Site selection (Phase 2 "where"; ADR 0014)

Candidates are drainage-network cells not within three cells of the map edge, with slope ≤ 15 %. Each is scored by a weighted sum of four memberships in [0, 1]:

- **upstream area** — a plateau in log10 hectares: 0 at 1 ha, 1 from 10 to 150 ha, 0 at 1 000 ha. A first version scored area monotonically and put every top site on the main river; a village pond is not a river dam;
- **flatness** — 1 on 0–3 % slope, falling to 0 at 15 % (an optimum at exactly 0 % would reward floodplains and DEM artefacts);
- **wetness** — normalised TWI;
- **impoundment efficiency** — the volume held behind a nominal 2 m rise (8-connected flood fill on the upstream side of a bund) divided by the pool footprint: a mean depth that rewards natural basins.

Weights come from a declared Saaty pairwise matrix by the principal eigenvector (AHP, Saaty 1980); the consistency ratio is computed with Saaty's random index and returned — CR = 0.004 for the default matrix; an intransitive matrix is rejected in a golden test. Non-maximum suppression at 200 m keeps the top-N distinct. The top site's catchment is delineated and returned with the ranking and every candidate's per-criterion scores, so the answer to "why here?" is in the response.

4.5 Rainfall (FR5)

Daily precipitation from Open-Meteo's ERA5-Land archive (1981–2025 for the sample's centroid), NASA POWER as fallback, both behind `Retry ∘ CircuitBreaker ∘ Cached` and a `FallbackChain` that records which provider answered. Statistics on complete calendar years only (incomplete years excluded, never scaled): mean, median, CV, **75 % dependable rainfall by the Weibull plotting position** $P = m/(n+1)$ (the design figure in Indian minor-irrigation practice), June–September share, IMD rainy days (≥ 2.5 mm), maximum 1-day, monthly normals, and the 25-year 1-day depth by a Gumbel EV1 fit (method of moments) on annual maxima.

4.6 Runoff (FR6)

Land cover from ESA WorldCover 2021 (10 m), read as a window straight from the public COG; topsoil texture from ISRIC SoilGrids → hydrologic soil group (USDA rule of thumb; default C with a warning when SoilGrids times out). The composite curve number is the area-weighted TR-55 value over the catchment (AMC II, Hawkins adjustment available). Three methods run on the daily series and report 75 % dependable annual runoff volumes:

- **SCS-CN** (USDA-SCS 1972; TR-55 1986): $Q = (P - I_a)^2 / (P - I_a + S)$, $S = 25400/CN - 254$ mm, $I_a = 0.2\ S$, per day, then summed — applied to an annual total it overestimates runoff more than threefold (asserted by a test);
- **runoff coefficient** (rational form, ASCE coefficients by land cover): the crude upper bracket;
- **Strange's table** (1928, Madras PWD) by daily rainfall and catchment condition.

The spread between them is reported, not hidden (188 % on the sample's top site); SCS-CN is the design figure.

4.7 Pond design (FR7; ADR 0016)

Target storage = 75 % dependable runoff × 0.6 harvest efficiency, clamped to [2 000, 50 000] m³ (a village pond, not a reservoir; when the cap binds the response says so). Geometry: an excavated inverted frustum, 2:1 side slopes, prismoidal volume, 0.5 m freeboard, bottom ≥ 5 m each way. **Depth is derived by cost**: a grid search over depth 1.5–3.5 m and aspect 1–2 solves each candidate's top dimensions for the target and picks the cheapest (₹160/m³ cut, ₹220/m³ bund, indicative 2024 bands), ties to the smaller water surface. A daily water balance over the record (inflow = SCS-CN runoff × area × 0.6; evaporation 0.7 × pan from a monthly climatology; seepage 2 mm/day; 15 % dead storage) yields the fill reliability — the share of years reaching ≥ 90 % — plus months with water and mean spill. The spillway is sized for the 25-year storm (IMD short-duration reduction, Kirpich time of concentration, rational peak, broad-crested weir). A bill of quantities and a confidence label decided by the worst input (assumed land cover or soil, cached rainfall, edge-limited catchment) complete the payload; the natural EAV curve of the site is reported alongside.

5. Software design and quality

- **Code quality:** `ruff` clean, `mypy --strict` on `domain/` and `engines/`, 203 tests, layering enforced by an executable test; every engine module's docstring names the algorithm and its citation. Coverage: **engines 94.3 %, domain 97.6 %, overall 86.2 %** (`docs/figures/p7-coverage.jpg`).
- **Test taxonomy:** golden tests with analytic answers (inclined plane, cone, V-valley, Y-valley, artificial pit, flat terrace, frustum hand calculation, SCS-CN hand calculation, monotonicity of runoff in rainfall), parser tests on the sample and on Z/ExtendedData/KMZ/ decoy variants, recorded-response tests for the rainfall adapters, resilience-decorator tests with injected failures, end-to-end API flows on the real sample with the in-memory adapters, the pysheds cross-check, and the saga compensation test.
- **Git hygiene:** conventional commits, one feature per commit, no secrets or generated binaries (COGs are regenerated by `make seed`), `.env.example` documents every variable.

6. System design and management (ADR 0019)

Async jobs with `202` + poll/WebSocket and real stage percentages; two bulkhead queues (a catchment returned in 0.6 s while a suitability job was running); idempotency keys; backpressure `429 + Retry-After`; RS256 JWT with viewer/planner/officer roles; a recommendation state machine whose transitions write a transactional outbox drained into an append-only audit log; request-correlation ids in every JSON log line; Prometheus metrics from the API and both workers with a provisioned Grafana dashboard; a leader-elected nightly rainfall refresh; an offline-first service worker (ADR 0018). Load test (Locust, 50 users, 60 s): 1 102 catchment submissions, POST p95 33 ms, end-to-end p95 560 ms, 0 HTTP failures. Chaos test: with the API container stopped, the reloaded page still shows the village, its layers, rainfall and sites from cache under an "Offline" badge (`docs/media/chaos-test.gif`); the test caught a real bug on its first run — a reverse proxy answers 502, which is a response, not a network error — now handled.

Scaling arithmetic and the distributed roadmap are in `docs/PLAN.md` Part 9; the bottleneck to national deployment is cadastral data and institutional trust, not compute.

7. Validation and results

Check	Method	Result
Interpolation	Inclined plane and cone contours, analytic elevations	RMS < 0.15 m (plane), < 0.25 m between rings (cone); summits above the top contour are flattened — a documented limitation
Sink filling	Synthetic pit; flat terrace	Pit filled exactly to its spill level, nothing else touched; every cell drains after +ε
Accumulation	Inclined plane (count of upslope cells); bowl	Exact match; bowl drains to its centre unfilled and to its rim filled
Delineation	V-valley: catchment = the rectangle upstream of the outlet; Y-valley: Strahler order 2	Exact
Independent implementation	pysheds (Bartos 2020) on the sample DEM, outlets snapped in both models	2.0 %, 3.4 % (Jaccard 0.98, 0.97) on the main outlets; 22.5 % (0.77) on a floodplain flat where the two flat-resolution schemes route differently — reported, not hidden (ADR 0015)
Pour-point sensitivity	Catchment area over a ±3-cell window around a channel cell	CV 212 % without snapping, 48 % with (`p2-pour-point-sensitivity.png`)
Stream threshold	Overlay on imagery	The 5 ha network reproduces the Sivnath and its main tributaries without hillside noise
Water mask	Sentinel-2 NDWI, three post-monsoon scenes	32 raw components → 20 water bodies, 9.0 % of the area; WorldCover 2021 independently says 8.1 %
AHP	Default matrix; perfectly consistent matrix; intransitive matrix	CR 0.004; 0.000; rejected (> 0.10)
Rainfall	Recorded ERA5-Land, Khapri	45 complete years; mean 1 313 mm, 75 % dependable 1 168 mm, monsoon share 89 %, 86 rainy days
Runoff	Hand calculation, CN 80, 50 mm day	Q = 13.8 mm; annual-total shortcut > 3× the daily sum
Design	Frustum hand calculation; optimiser feasibility; water balance on synthetic seasons	Exact; cheapest feasible design chosen; small pond fills every year, oversized pond never
Existing ponds	OSM's two mapped tanks inside the extent, designed at their centroids	Local runoff is a tenth of their capacity (canal-fed tanks in the Durg command area); natural impoundment at +2 m within −40 % of the larger tank's capacity (`p3-existing-pond-comparison.md`)
Load	Locust 50 users / 60 s	POST p95 33 ms, end-to-end p95 560 ms

Sample-map headline numbers (Khapri, Durg, Chhattisgarh, EPSG:32644, 830 ha): top site drains 38.25 ha (±26 %); 75 % dependable rainfall 1 168 mm; CN 88 (HSG C); SCS-CN 99 k m³ (rational 244 k, Strange 18 k); design 50 000 m³ (capped; harvestable 59.5 k), 3.5 m deep, 126 × 126 m top, fills in 100 % of years, ₹97 lakh indicative, confidence *low* (soil assumed); 27 ha of eligible land in 16 patches.

8. Limitations and uncertainty

The DEM is interpolated from contours that are themselves interpolated from ~30 m SRTM (±6 m relative): relief below about 5 m is not real, single-cell extremes are lost, and the sample's summit and valley floor sit 2–3 m inside the contour range. Every storage figure therefore carries ±20 % or worse and is planning-grade, not survey-grade. Ownership is unknown without a cadastral layer and is never assumed. Reanalysis rainfall is ±15 % against gauges. The three runoff methods disagree by up to 2×; the design uses SCS-CN and says so. SoilGrids is slow and the default soil group is a stated assumption. The ML scoring path is deferred by the plan's own fallback: two mapped tanks are not a training set (ADR 0017).

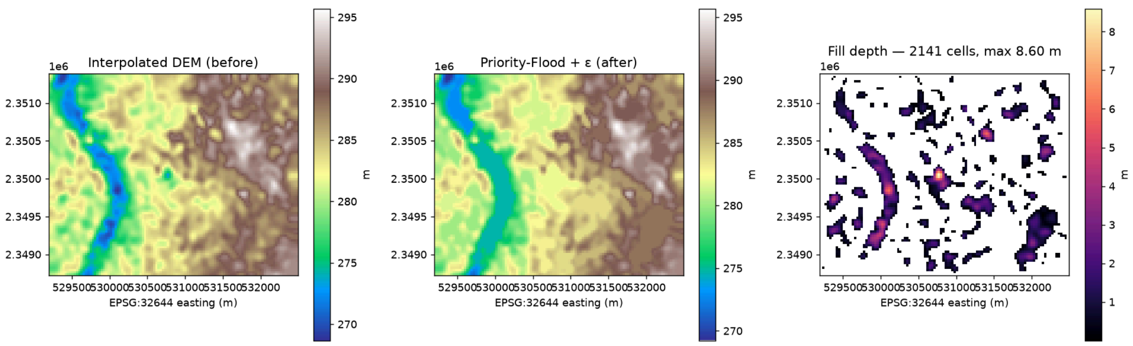
9. Future work

Ground-truth against 20–30 existing ponds in one block; a cadastral import with ownership *class* only (DPDP Act 2023); local CN calibration; the provider-DEM adapter (Copernicus GLO-30) for villages without a contour map; the ML/SHAP path once labels exist; the distributed roadmap in `docs/PLAN.md` Part 9.

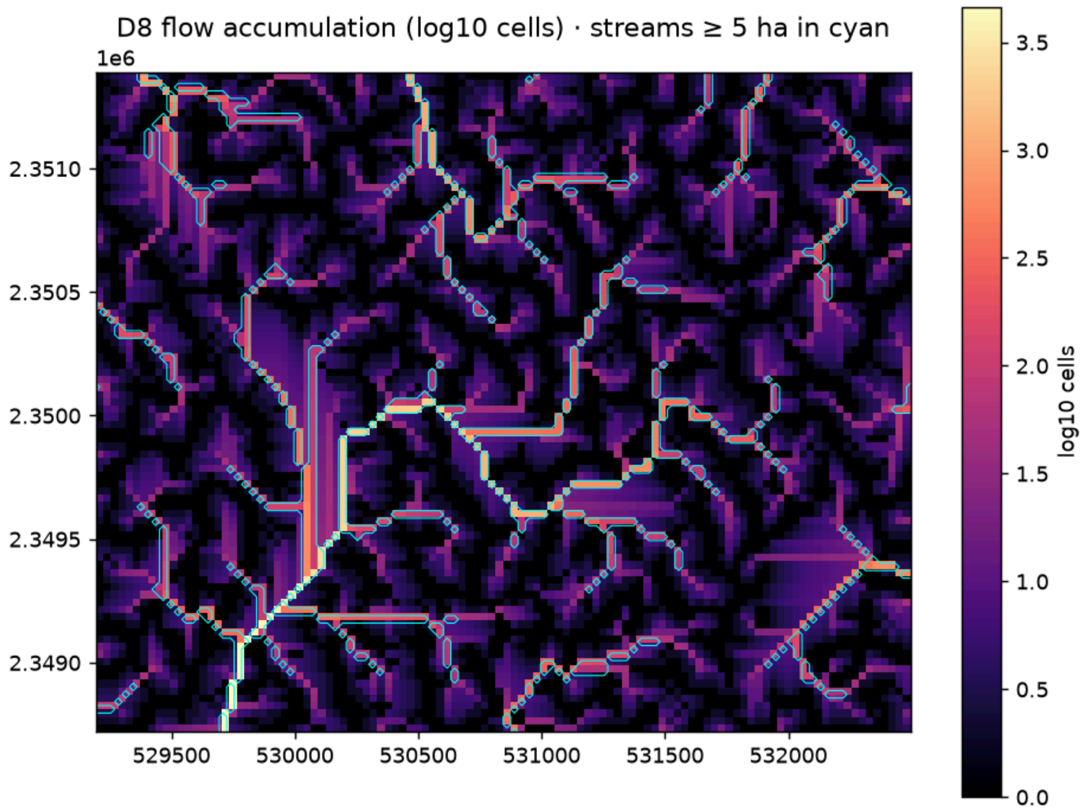
10. Use of AI tools

Claude Code (Anthropic) was used throughout for drafting code, tests and documentation, for refactoring and for debugging, under the author's direction and review, as the assignment's LLM policy permits. Every design decision is recorded with its reasoning and rejected alternatives in `docs/adr/` (19 records) and the decision log in `docs/PROGRESS.md`; every algorithm is named and cited in its module docstring; the author can explain and justify each component live.

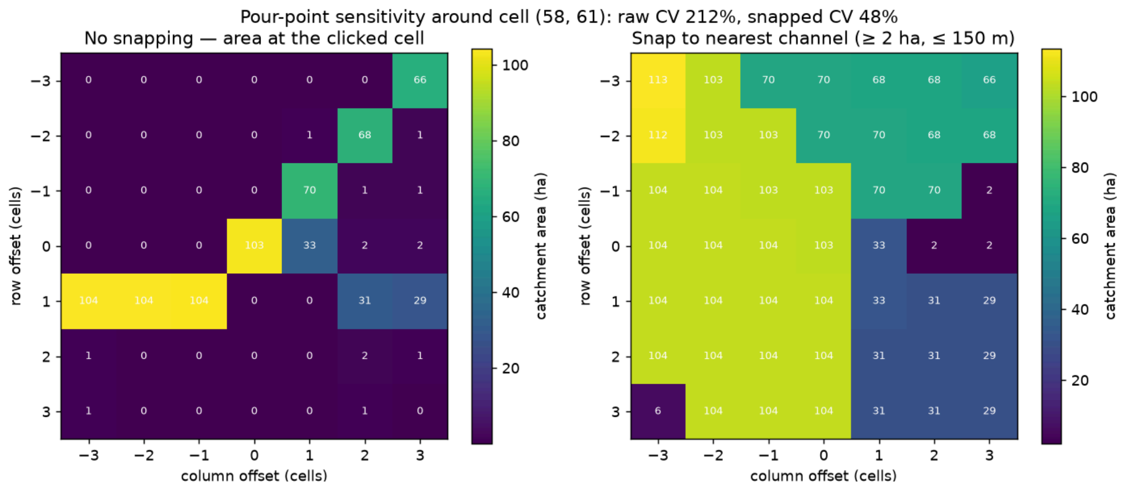
Appendix A — Figures



Sink filling on the sample DEM: 2 141 cells filled, maximum 8.6 m, before/after



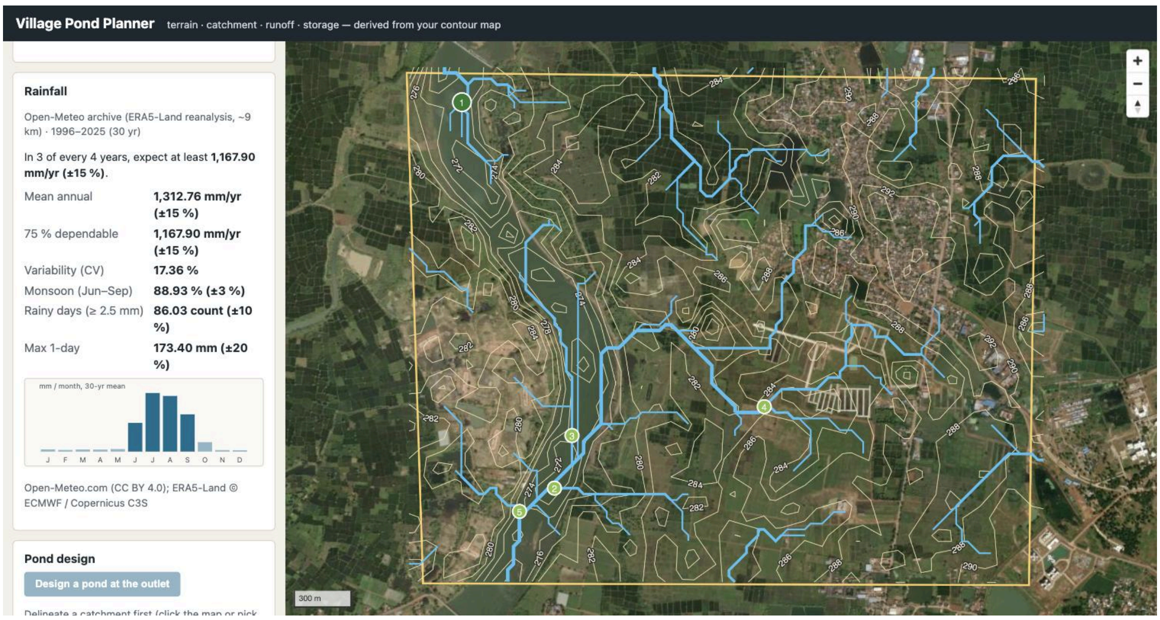
D8 flow accumulation (log scale) with the 5 ha stream network



Pour-point sensitivity: catchment area around a channel cell, raw vs snapped

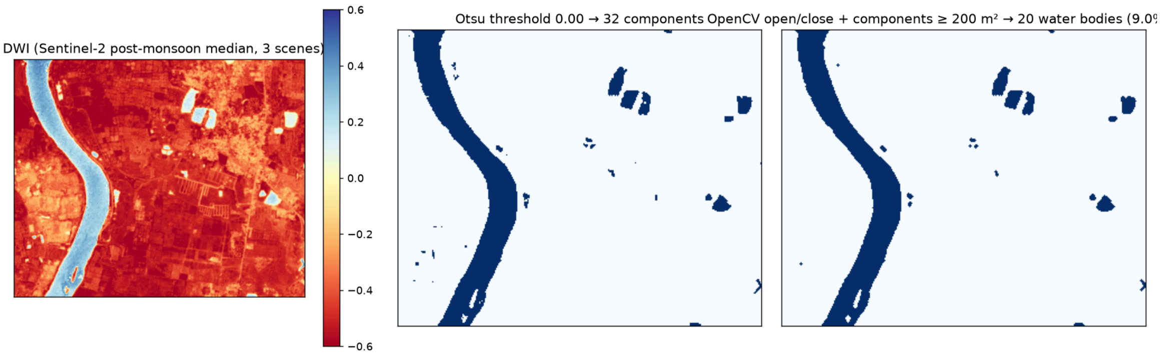
Modelled streams over satellite imagery — the Sivnath and its tributaries

Click-to-catchment in the browser: snapped outlet, snap distance, polygon, area with its band

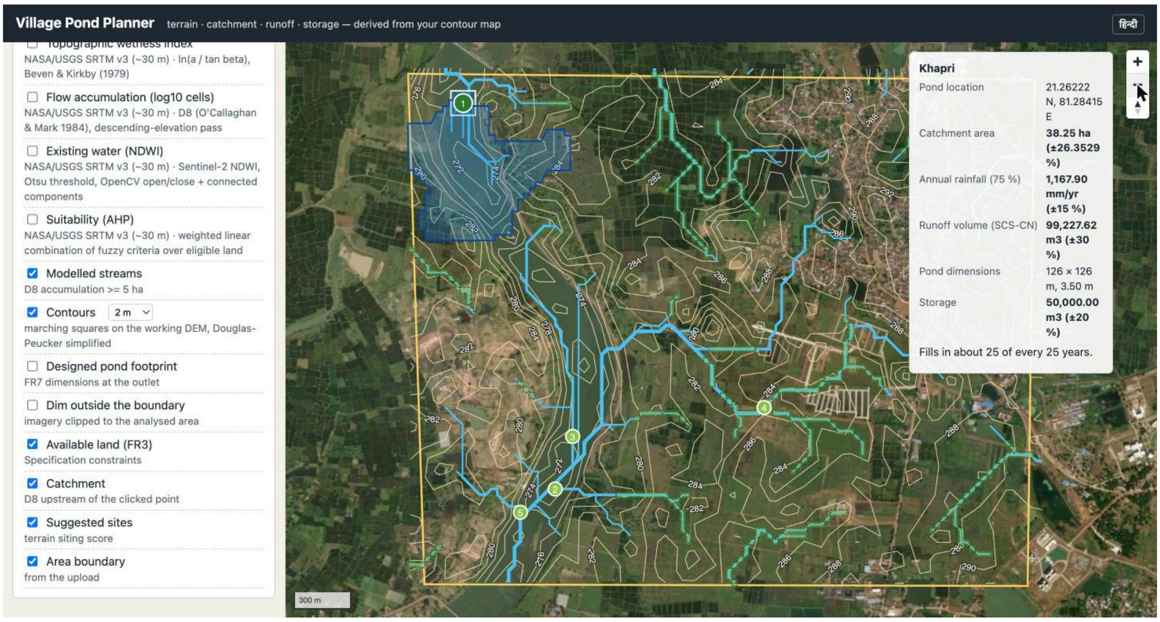


Rainfall panel: 45 years of ERA5-Land, 75 % dependable year, monthly normals

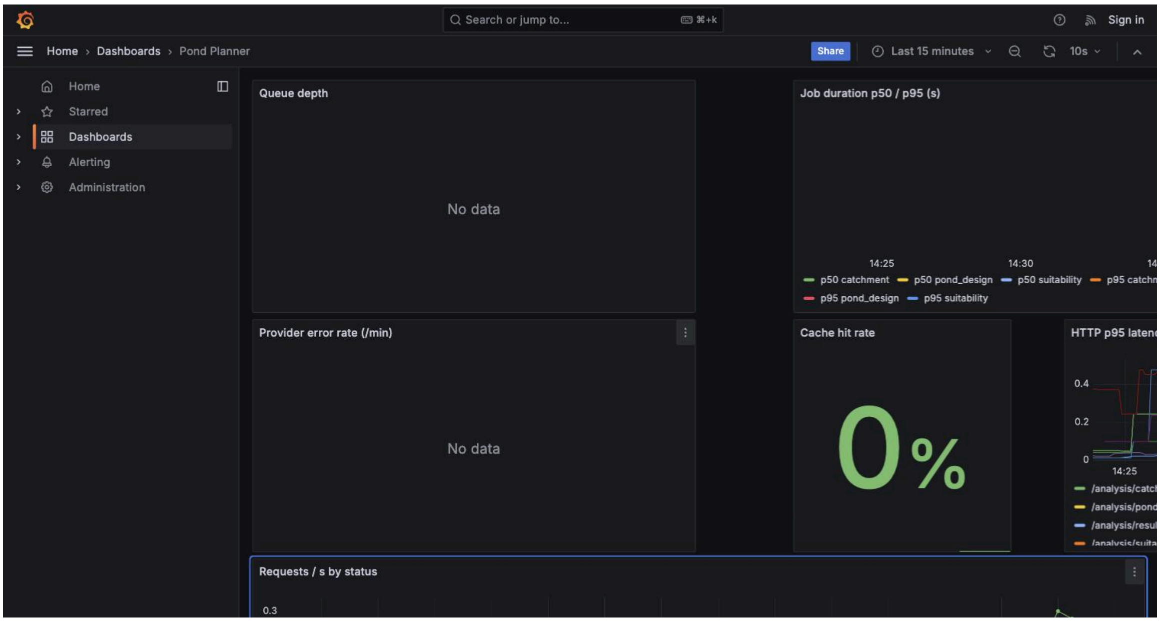
Pond design panel: dimensions, EAV curve, three runoff methods, fill reliability



NDWI water mask: Otsu threshold, OpenCV morphology, connected components



All FR8 overlays on one map



Grafana dashboard during the load test

Offline: results served from cache with the API container stopped

Coverage report: 86%

Files Functions Classes

coverage.py v7.15.4, created at 2026-08-26 14:42 +0530

app/engines

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File	statements	missing	excluded	coverage
app/engines/__init__.py	0	0	0	100%
app/engines/design/__init__.py	0	0	0	100%
app/engines/design/eav.py	48	0	0	100%
app/engines/design/geometry.py	64	0	0	100%
app/engines/design/optimiser.py	37	1	0	97%
app/engines/design/spillway.py	19	0	0	100%
app/engines/design/water_balance.py	59	1	0	98%
app/engines/hydrology/__init__.py	0	0	0	100%
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app/engines/rainfall/__init__.py	0	0	0	100%
app/engines/rainfall/service.py	38	5	0	87%
app/engines/rainfall/statistics.py	80	4	0	95%
app/engines/runoff/__init__.py	0	0	0	100%
app/engines/runoff/curve_number.py	38	2	0	95%
app/engines/runoff/methods.py	77	1	9	99%
app/engines/suitability/__init__.py	0	0	0	100%
app/engines/suitability/ahp.py	35	2	0	94%
app/engines/suitability/constraints.py	110	9	0	92%
app/engines/suitability/water_mask.py	44	0	0	100%

Coverage report

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